

SOFTWARE REQUIREMENTS SPECIFICATION

ALUMNI WEB APPLICATION

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01. Project Overview

The Alumni Portal is a comprehensive web-based application designed to connect past students, current professionals, and educational institutions including colleges, schools, and offices. This platform serves as a unified digital bridge that enables alumni to maintain lifelong connections with their alma mater and fellow graduates.

1.1 Purpose

The Alumni Web Application aims to create a centralized platform where alumni can register, share career updates, network with peers, and stay engaged with their institution. The system facilitates seamless communication between institutions and their graduates while providing administrators with powerful tools to manage and showcase alumni achievements.

1.2 Key Objectives

- Build a unified networking platform for alumni across different batches and departments
- Enable institutions to effectively manage and showcase alumni achievements and success stories
- Provide secure authentication mechanisms for both administrators and users
- Implement advanced search and filtering capabilities for finding alumni by profession, batch, organization, or location
- Create a responsive and user-friendly interface accessible across all devices
- Establish a secure and scalable foundation for future feature enhancements

1.3 Scope

The Alumni Portal encompasses three primary user roles with distinct functionalities. Administrators have comprehensive control over alumni data with full CRUD operations, user management, and approval workflows. Alumni users can view, search, and connect with fellow graduates through a secure login system. The public interface showcases alumni achievements and provides registration capabilities through institution-linked forms. The application supports multiple institutions, each with customized branding and dedicated data management.

02. Website Features - Customised Landing Modules

2.1 Home Page Module

The home page serves as the main entry point featuring a visually appealing layout with the following components:

- Dynamic Header with navigation menu (Home, Register, User Login, Dashboard, Logout)
- Hero section showcasing 'Our Distinguished Alumni' with engaging visuals
- Alumni network counter displaying total registered members
- Alumni profile cards with carousel navigation showing key details
- Employment status badges (Studying, Employed, Self-Employed)
- Profile information including name, profession, organization, and batch year
- Comprehensive footer with contact information and quick links

2.2 Registration Module

A comprehensive registration system designed for alumni including students, officemates, and professionals from various institutions:

- User-friendly multi-step registration form with clear field labels
- Photo upload functionality with file size validation (JPG/PNG/GIF supported)
- Login credentials section for username and email address creation
- Personal information fields (full name, contact details, current location)
- Professional details section (current position, company/organization)
- Academic information (institution name, department, graduation year)
- Employment status selection (Studying, Employed, Self-Employed)
- Password creation with confirmation field and strength indicator
- Real-time form validation with helpful error messages
- Required field indicators with asterisk notation

2.3 Admin Dashboard Module

A powerful administrative interface providing comprehensive alumni database management:

- Dashboard overview with key statistics and metrics
- Add New Alumni button with streamlined profile creation workflow
- Manage Alumni section with view, edit, and delete capabilities
- Manage Users functionality for account creation and permissions
- Statistics Overview displaying total alumni count by employment status
- Visual analytics with icons showing Employed, Self-Employed, and Studying counts
- Search and filter tools for efficient data management
- Bulk operations support for data import/export
- User credential management system for creating and resetting login passwords

03. Website Features - Alumni Portal

3.1 User Authentication System

Feature	Description
Admin Login	Secure authentication with username/email and password for administrative access
User Login	Alumni member login with view-only access to alumni directory
Password Security	Encrypted password storage using bcrypt with salt rounds for enhanced security
Session Management	JWT token-based authentication with configurable expiration times

3.2 Alumni Directory Features

Feature	Description
Profile Display	Comprehensive alumni profiles with photo, profession, organization, batch year, department, and location
Advanced Search	Multi-criteria search by name, profession, company, batch year, department, or location
Filter Options	Filter by employment status (Studying, Employed, Self-Employed), batch year, or department
Contact Feature	Direct contact options through email or integrated messaging system

3.3 Data Management Capabilities

Administrators possess comprehensive control over the alumni database:

- Create new alumni profiles with complete information capture
- View detailed alumni information with full profile history
- Update existing records including personal, academic, and professional details
- Delete alumni records with confirmation safeguards
- Approve or reject alumni registration requests
- Export alumni data in CSV/Excel format for reporting
- Import bulk alumni data with validation and error handling
- Create and manage user login credentials for alumni access
- Reset passwords and manage user account permissions

04. Website Navigation Structure: Front End - English

4.1 Public Interface Navigation

Page	URL Path	Description
Home	/home or /	Landing page displaying featured alumni profiles and network statistics
Register	/register	Alumni registration form for new member enrollment
User Login	/user/login	Login portal for registered alumni members
About Us	/about	Information about the institution and alumni network
Contact	/contact	Contact form and institution contact details

4.2 User Interface Navigation (Alumni Members)

Page	URL Path	Description
User Dashboard	/user/dashboard	Personal dashboard for logged-in alumni
Alumni Directory	/admin/manage	View-only access to complete alumni directory
Search Alumni	/ admin/manage	Advanced search and filter interface
My Profile	/user/profile	View personal profile and update password

05. Administrative Module

5.1 Admin Authentication

Secure administrator login with multi-factor authentication support. Admin credentials are managed through encrypted storage with role-based access control ensuring only authorized personnel can access administrative functions.

5.2 Dashboard Overview

The admin dashboard provides a comprehensive overview with:

- Real-time statistics on total alumni count
- Employment status breakdown (Employed, Self-Employed, Studying)
- Recent registration requests pending approval
- Quick action buttons for common tasks
- System notifications and alerts

5.3 Alumni Management Functions

5.3.1 Add New Alumni

Administrators can create new alumni profiles with comprehensive data input including personal details, academic information, professional background, and contact information. The system validates all inputs and ensures data integrity.

5.3.2 Manage Alumni

Complete CRUD operations for alumni records including view detailed profiles, edit any field with validation, delete records with confirmation, approve or reject registration requests, and bulk operations for efficiency.

5.3.3 Manage Users

User account management capabilities including create login credentials for alumni members, reset passwords when requested, manage user permissions and access levels, deactivate or suspend user accounts, and track login activity and user engagement.

5.4 Search and Filter Tools

Advanced search capabilities for administrators:

- Full-text search across all alumni fields
- Filter by employment status, batch year, department, location
- Sort results by various criteria (name, date, status)
- Export filtered results to Excel/CSV
- Save custom filter configurations

06. Website Navigation Structure: Back End - Admin Section

6.1 Admin Panel Navigation

Module	URL Path	Access Level
Admin Login	/admin/login	Public (authentication required)
Dashboard	/admin/dashboard	Admin Only
Add Alumni	/admin/alumni/add	Admin Only
Manage Alumni	/admin/alumni/manage	Admin Only
Edit Alumni	/admin/alumni/edit/:id	Admin Only
Delete Alumni	/admin/alumni/delete/:id	Admin Only
Manage Users	/admin/users/manage	Admin Only
Create User	/admin/users/create	Admin Only
Reports	/admin/reports	Admin Only
Settings	/admin/settings	Admin Only

6.2 API Endpoints (Backend)

Method	Endpoint	Description
POST	/api/admin/login	Admin authentication endpoint
GET	/api/alumni	Retrieve all alumni records
GET	/api/alumni/:id	Retrieve single alumni record
POST	/api/alumni	Create new alumni record
PUT	/api/alumni/:id	Update alumni record
DELETE	/api/alumni/:id	Delete alumni record
POST	/api/users/create	Create user login credentials
POST	/api/search	Search and filter alumni

07. Domain, Hosting & Other Requirements

7.1 Domain Requirements

Domain Name	Custom domain (e.g., alumni.institution.edu or alumninet.com)
SSL Certificate	Required for HTTPS (Let's Encrypt or commercial SSL)
DNS Management	Access to DNS settings for configuration

7.2 Hosting Specifications

Component	Specification
Server Type	VPS or Cloud Server (AWS, Digital Ocean, Vercel, or Railway)
CPU	Minimum 2 vCPU cores (recommended 4 cores for production)
RAM	Minimum 4GB (recommended 8GB for optimal performance)
Storage	50GB SSD storage minimum (scalable based on alumni count)
Bandwidth	Minimum 1TB/month data transfer
Operating System	Ubuntu 20.04 LTS or higher, CentOS 8, or similar Linux distribution
Database	MySQL 8.0 or higher (dedicated or managed service)

7.3 Additional Requirements

- Email service for notifications (SendGrid, AWS SES, or SMTP server)
- CDN for static assets and image delivery (Cloudflare or AWS CloudFront)
- Backup solution with daily automated backups
- Monitoring and alerting system (e.g., PM2, New Relic, or DataDog)
- File storage for alumni photos (local or cloud storage like AWS S3)

08. Technical Proposal

8.1 Technology Stack

8.1.1 Frontend Technologies

Framework	React.js with Vite build tool
UI Components	Material-UI or Tailwind CSS for responsive design
State Management	React Context API or Redux for complex state management
Routing	React Router v6 for client-side navigation
Form Handling	React Hook Form with Yup validation

8.1.2 Backend Technologies

Runtime	Node.js (LTS version 18.x or higher)
Framework	Express.js for RESTful API development
Authentication	JWT (JSON Web Tokens) for secure session management
Password Encryption	bcrypt with salt rounds for password hashing
File Upload	Multer middleware for handling multipart/form-data
Validation	express-validator for input sanitization and validation

8.1.3 Database

Database System	MySQL 8.0 relational database
ORM/Query Builder	Sequelize ORM or raw MySQL2 driver for direct queries
Connection Pooling	Configured for optimal performance with 10-20 connections

8.2 Database Schema Design

8.2.1 Alumni Table

Primary table storing comprehensive alumni information:

- id (INT, PRIMARY KEY, AUTO_INCREMENT)
- name (VARCHAR 255, NOT NULL)
- email (VARCHAR 255, UNIQUE, NOT NULL)
- phone (VARCHAR 20)
- profession (VARCHAR 100)
- company (VARCHAR 255)
- college (VARCHAR 255)
- department (VARCHAR 100)
- batch_year (INT)

- location (VARCHAR 255)
- status (ENUM: 'Studying', 'Employed', 'Self-Employed')
- photo (VARCHAR 500 - file path)
- created_at (TIMESTAMP, DEFAULT CURRENT_TIMESTAMP)
- updated_at (TIMESTAMP, ON UPDATE CURRENT_TIMESTAMP)

8.2.2 Users Table

Authentication and user management table:

- id (INT, PRIMARY KEY, AUTO_INCREMENT)
- username (VARCHAR 100, UNIQUE, NOT NULL)
- email (VARCHAR 255, UNIQUE, NOT NULL)
- password_hash (VARCHAR 255, NOT NULL)
- role (ENUM: 'admin', 'user', DEFAULT 'user')
- alumni_id (INT, FOREIGN KEY references alumni.id)
- last_login (TIMESTAMP)
- is_active (BOOLEAN, DEFAULT TRUE)
- created_at (TIMESTAMP, DEFAULT CURRENT_TIMESTAMP)

8.3 Security Measures

- Password hashing with bcrypt (minimum 10 salt rounds)
- JWT token-based authentication with secure HTTP-only cookies
- CORS (Cross-Origin Resource Sharing) configuration
- SQL injection prevention through parameterized queries
- XSS (Cross-Site Scripting) protection with input sanitization
- CSRF (Cross-Site Request Forgery) tokens for form submissions
- Rate limiting to prevent brute force attacks
- HTTPS enforcement with SSL/TLS certificate
- File upload validation (type, size, malware scanning)
- Environment variables for sensitive configuration

09. Proposal Confidentiality & Disclosure Clause

This Software Requirements Specification document contains confidential and proprietary information belonging to Sunsys Technologies Pvt Ltd. The information contained herein is provided solely for the purpose of evaluating our proposed Alumni Web Application solution and must be treated with strict confidentiality.

9.1 Confidential Information

Confidential information includes but is not limited to:

- Technical specifications and architecture details
- Development methodologies and processes
- Pricing information and cost estimates
- Implementation timelines and resource allocation
- Proprietary software solutions and algorithms
- Any other technical or business information marked as confidential

9.2 Recipient Obligations

The recipient of this document agrees to:

- Maintain confidentiality of all information contained in this document
- Not disclose any information to third parties without written consent
- Use the information solely for evaluation purposes
- Return or destroy all copies upon request or project completion
- Limit access to authorized personnel with legitimate need-to-know

9.3 Intellectual Property

All intellectual property rights including but not limited to software code, architecture designs, documentation, and methodologies described in this proposal remain the exclusive property of Sunsys Technologies Pvt Ltd. No license or rights are granted except as explicitly stated in a signed agreement.

9.4 Duration

The confidentiality obligations shall remain in effect for a period of three (3) years from the date of disclosure, or until the information becomes publicly available through no fault of the recipient, whichever occurs first.

10. Development Sprints & Milestones

10.1 Project Timeline Overview

The Alumni Web Application will be developed using an Agile methodology with 2-week sprint cycles. The complete project is estimated to be completed in 12-14 weeks, divided into 6-7 sprints with clear deliverables and milestones.

10.2 Sprint Breakdown

Sprint 1: Project Setup & Database Design (Weeks 1-2)

Deliverables	• Project repository setup and version control • Development environment configuration • Database schema design and implementation • Basic Express.js server setup with MySQL connection • Project documentation structure
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Sprint 2: Authentication System (Weeks 3-4)

Deliverables	• Admin login API with JWT authentication • User login API for alumni members • Password encryption with bcrypt • Session management and token refresh • Protected route middleware
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Sprint 3: Frontend Foundation & Public Pages (Weeks 5-6)

Deliverables	• React Vite project setup with routing • Responsive home page with header and footer • Alumni profile card component with carousel • Registration form with validation • Login pages for admin and users
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Sprint 4: Alumni Management Backend (Weeks 7-8)

Deliverables	• CRUD APIs for alumni management • File upload API with Multer integration • Search and filter API endpoints • User management APIs • Data validation and error handling
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Sprint 5: Admin Dashboard (Weeks 9-10)

Deliverables	• Admin dashboard with statistics overview • Add new alumni form and functionality • Manage alumni table with edit/delete options • User management interface • Search and filter implementation
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Sprint 6: User Interface & Alumni Directory (Weeks 11-12)

Deliverables	• User dashboard for logged-in alumni • Alumni directory with view-only access • Advanced search and filter interface • Profile view page • Responsive design optimization
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Sprint 7: Testing, Deployment & Documentation (Weeks 13-14)

Deliverables	• Comprehensive testing (unit, integration, E2E) • Bug fixes and performance optimization • Production deployment setup • User documentation and admin guide • Training materials and handover
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10.3 Key Milestones

Milestone	Timeline	Deliverable
M1: Project Kickoff	Week 1	Project charter and requirements sign-off
M2: Backend Core	Week 4	Functional authentication and database
M3: Frontend MVP	Week 6	Working home page and registration
M4: Admin Features	Week 10	Complete admin dashboard with CRUD
M5: User Features	Week 12	Alumni directory with search functionality
M6: Go Live	Week 14	Production deployment and launch

10.4 Post-Launch Support

Following successful deployment, Sunsys Technologies Pvt Ltd will provide:

- 30 days of free post-launch support and bug fixes
- Administrator training sessions (2-3 sessions)
- Comprehensive documentation and user guides
- Email and phone support during business hours
- Optional maintenance and enhancement contract available

11. Conclusion

The Alumni Web Application proposed by Sunsys Technologies Pvt Ltd represents a comprehensive solution designed to strengthen the connection between educational institutions and their graduates. Built with modern technologies including React Vite, Node.js, and MySQL, this platform offers robust functionality for both administrators and alumni members.

Our systematic approach ensures successful delivery through well-defined sprints, clear milestones, and comprehensive testing. The platform's security features, scalable architecture, and user-friendly interface position it as a long-term asset for alumni engagement and institutional networking.

We are committed to delivering a high-quality product that meets all specified requirements while maintaining the flexibility to accommodate future enhancements. Our post-launch support ensures smooth operation and provides administrators with the knowledge and tools needed for effective alumni management.

Thank you for considering Sunsys Technologies Pvt Ltd for your Alumni Web Application project.

Contact Information

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For any queries or additional information, please feel free to contact us.

SYSTEM FLOWCHARTS

ALUMNI WEB APPLICATION

Process Flows & System Architecture

Prepared by:

Sunsys Technologies Pvt Ltd

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01. Introduction

This document provides comprehensive flowcharts and process diagrams for the Alumni Web Application. Each flowchart illustrates the logical flow of operations, decision points, and data movement throughout the system.

1.1 Purpose

These flowcharts serve multiple purposes including system documentation, developer reference, training materials for administrators, and quality assurance testing guidelines. They provide a visual representation of how users interact with the system and how data flows through various components.

1.2 Flowchart Symbols Legend

Symbol	Name	Description
()	Start/End	Represents the beginning or end of a process
[]	Process	Represents a process, action, or operation
{ }	Decision	Represents a decision point with Yes/No or multiple options
→	Flow Direction	Indicates the direction of process flow
(D)	Database	Represents database operations or storage

02. Main System Flowchart

The main system flowchart illustrates the complete user journey through the Alumni Web Application, showing all major entry points, user roles, and their respective workflows.

2.1 Entry Points

START	Alumni Web Application Launch
↓	
Entry Point?	Three possible entry points: 1. Public User → Home Page 2. Admin → Admin Login 3. Alumni Member → User Login

2.2 Public User Flow

- User lands on Home Page
- Views featured alumni profiles in carousel
- Sees alumni network statistics
- Clicks Register button
- Fills registration form with personal details
- Uploads photo (max 5MB, JPG/PNG/GIF)
- Creates username and email
- Sets secure password
- Submits registration
- Status set to 'Pending Admin Approval'
- Waits for admin review and approval

2.3 Admin User Flow

2.3.1 Login Process

Step 1	Admin navigates to Admin Login page
Step 2	Enters admin credentials (username/email and password)
Decision	Valid credentials? • If NO → Show error message → Return to login • If YES → Generate JWT token → Redirect to Admin Dashboard

2.3.2 Admin Dashboard Actions

Once logged in, admin can choose from the following actions:

Action	Process	Outcome
View Statistics	Display dashboard with counts	Shows total, employed, self-employed, studying
Add Alumni	Open form, enter details, upload photo	New alumni record saved to database

Action	Process	Outcome
Manage Alumni	View list, search, edit or delete	Alumni records updated or removed
Manage Users	Create login, reset password	User accounts created or modified
Approve Registrations	Review pending, approve/reject	Account created or registration rejected

2.4 Alumni User Flow

2.4.1 Login Process

- Alumni member clicks User Login
- Enters credentials (username/email and password)
- System validates credentials
- If invalid → Show error → Return to login
- If valid → Generate JWT token → Redirect to User Dashboard

2.4.2 User Dashboard Actions

- **View Alumni Directory:**

Browse all approved alumni profiles with view-only access

- **Search Alumni:**

Use advanced search with filters for name, profession, company, batch, department, or location

- **My Profile:**

View personal profile information and change password

- **Contact Alumni:**

Send messages or emails to connect with fellow alumni

03. Authentication & Security Flowchart

This flowchart details the complete authentication process including security measures, token management, and role-based access control.

3.1 Authentication Process Overview

Step	Process	Details
1	User Visits Site	System checks for existing authentication token
2	Authentication Check	If authenticated → Check role → Route to dashboard If not authenticated → Show login options
3	Enter Credentials	User enters username/email and password in login form
4	Rate Limit Check	System checks for excessive login attempts If too many attempts → Block request (429 error) → Wait cooldown If OK → Proceed to validation
5	Validate Credentials	Query database for username/email If user not found → Show error → Return to login If user found → Proceed to password verification
6	Password Verification	Use bcrypt to compare entered password with stored hash If password incorrect → Show error → Return to login If password correct → Check account status
7	Account Status Check	Verify account is active (is_active = true) If inactive → Show inactive message → Return to login If active → Generate JWT token
8	Generate JWT Token	Create JWT token with user ID, role, and expiration time Token payload: {userId, role, exp: 24h} Sign token with secret key
9	Set Session Cookie	Store JWT token in HTTP-only cookie Set secure flag for HTTPS transmission Update last_login timestamp in database
10	Redirect to Dashboard	Check user role from token If role = admin → Redirect to Admin Dashboard If role = user → Redirect to User Dashboard

3.2 Security Measures

Security Feature	Implementation
Password Hashing	bcrypt algorithm with 10+ salt rounds Passwords never stored in plain text One-way hashing prevents password recovery
JWT Token Authentication	Stateless authentication mechanism Token expires after 24 hours Signed with secret key to prevent tampering Includes user ID and role for authorization
Rate Limiting	Maximum 5 failed login attempts per 15 minutes Prevents brute force attacks Temporary account lockout after threshold
HTTP-Only Cookies	Tokens stored in HTTP-only cookies Not accessible via JavaScript (XSS protection) Secure flag ensures HTTPS-only transmission
HTTPS/TLS Encryption	All data transmitted over encrypted HTTPS SSL/TLS certificate required Protects credentials during transmission
Role-Based Access Control	Admin role: Full CRUD access to all features User role: View-only access to alumni directory Middleware validates role on every request

3.3 Session Management Flow

- User makes authenticated request
- Middleware extracts JWT token from cookie
- Verify token signature and expiration
- If token expired → Check refresh token availability
- If refresh available → Issue new token → Continue session
- If no refresh → Force logout → Redirect to login
- If token valid → Extract user data → Allow request
- Check role permissions for requested action
- If authorized → Process request → Return response
- If unauthorized → Return 403 Forbidden error

3.4 Registration and Approval Flow

Step 1	User fills registration form with personal and professional details
Step 2	Uploads photo (validated for size max 5MB and type JPG/PNG/GIF)
Step 3	Creates password (bcrypt hashes password with 10 salt rounds)
Step 4	System saves registration with status 'Pending Approval' to database
Step 5	Admin reviews registration in approval queue
Decision	Admin Decision: If APPROVED:

	• Create user account in users table • Link user account to alumni record (alumni_id foreign key) • Send email with login credentials • User can now login with credentials If REJECTED: • Send rejection notification email • User can re-register if desired
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3.5 Logout Process

4. User clicks logout button
5. System clears JWT token from memory
6. Delete session cookie
7. Clear any cached user data
8. Redirect user to home page
9. User must login again to access authenticated features

04. System Architecture & Data Flow

This section describes the three-tier architecture of the Alumni Web Application, showing how data flows between the client, server, and database layers.

4.1 Three-Tier Architecture Overview

Layer	Technology	Responsibilities
Client Layer (Presentation)	React with Vite Material-UI / Tailwind CSS React Router	- User interface rendering - Client-side routing - Form validation - State management
API Layer (Business Logic)	Node.js Express.js framework JWT authentication Multer (file uploads)	- RESTful API endpoints - Business logic processing - Authentication & authorization - Data validation
Database Layer (Data Storage)	MySQL 8.0 Sequelize ORM Connection pooling	- Data persistence - Query execution - Transaction management - Data integrity

4.2 Request-Response Data Flow

Example: Adding a New Alumni Record

10. CLIENT:

Admin fills out 'Add Alumni' form in React interface
 Uploads alumni photo
 Client validates form data (required fields, email format, etc.)
 Sends POST request to /api/alumni with form data and photo

11. API LAYER:

Express Router receives POST request
 Rate Limiter checks request frequency
 Auth Middleware verifies JWT token and admin role
 Validation Middleware sanitizes and validates input data
 Multer processes photo upload and saves to storage
 Alumni Controller receives validated data

12. DATABASE LAYER:

Controller executes INSERT query on alumni table
 MySQL validates data against schema constraints
 Record saved with auto-generated ID
 Database returns success response with new record ID

13. RESPONSE:

Controller formats success response
 Express sends JSON response to client
 Client receives response and shows success message
 Dashboard statistics updated automatically
 UI redirects to alumni list showing new record

05. Admin CRUD Operations Flowchart

This flowchart details all Create, Read, Update, and Delete operations available to administrators for managing the alumni database.

5.1 CREATE Operation

Process Flow for Adding New Alumni:

Start	Admin clicks 'Add New Alumni' button from dashboard
Step 1	System displays Add Alumni form with empty fields
Step 2	Admin fills required fields: • Name (required) • Email (required, unique) • Phone number • Profession • Current company/organization • College name • Department • Batch year (graduation year) • Current location • Employment status (Studying/Employed/Self-Employed)
Step 3	Admin uploads alumni photo
Validation	Photo Validation: • Check file size (max 5MB) • Check file type (JPG, PNG, GIF only) • If invalid → Show error message → Return to upload step • If valid → Proceed to form validation
Validation	Form Validation: • Check all required fields are filled • Validate email format and uniqueness • Validate phone number format • If validation fails → Show specific error messages → Return to form • If validation passes → Proceed to save
Step 4	System saves photo to file storage (local filesystem or AWS S3)
Step 5	Execute database INSERT query: INSERT INTO alumni (name, email, phone, profession, company, college, department, batch_year, location, status, photo, created_at) VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, NOW())
Success	Display success message: 'Alumni record created successfully' Update dashboard statistics (total count, status count) Redirect to alumni list or dashboard

5.2 READ Operations

5.2.1 View All Alumni

- Admin clicks 'Manage Alumni' from dashboard
- System executes: `SELECT * FROM alumni ORDER BY created_at DESC`
- Display alumni list in table format with key columns
- Show action buttons (View, Edit, Delete) for each record
- Display pagination controls if records exceed page limit

5.2.2 View Single Alumni Details

- Admin clicks 'View' button on alumni record
- System executes: `SELECT * FROM alumni WHERE id = ?`
- Display complete profile with all fields
- Show alumni photo
- Provide options to Edit or Delete from detail view

5.2.3 Search and Filter

- Admin opens search interface
- Selects search criteria (name, profession, company, batch, department, status, location)
- System builds dynamic SQL query with WHERE clauses
- Executes query: `SELECT * FROM alumni WHERE [conditions] ORDER BY name ASC`
- Display filtered results in table
- If no results → Show 'No alumni found' message
- Provide export to CSV/Excel option

5.3 UPDATE Operation

Start	Admin selects alumni to edit from list or detail view
Step 1	Fetch current alumni data: <code>SELECT * FROM alumni WHERE id = ?</code>
Step 2	Populate edit form with existing data
Step 3	Admin modifies desired fields
Decision	Change photo? If YES: - Admin uploads new photo - Validate new photo (size, type) - If invalid → Show error → Return to upload - If valid → Delete old photo from storage - Save new photo to storage - Update photo path in form data If NO: - Keep existing photo path - Proceed to validation
Validation	Validate all modified fields If validation fails → Show errors → Return to form If validation passes → Proceed to update
Step 4	Execute UPDATE query:

	UPDATE alumni SET name=?, email=?, phone=?, profession=?, company=?, college=?, department=?, batch_year=?, location=?, status=?, photo=?, updated_at=NOW() WHERE id=?
Success	Display success message: 'Alumni record updated successfully' Update dashboard statistics if status changed Redirect to alumni list or detail view

5.4 DELETE Operation

Start	Admin selects alumni to delete and clicks 'Delete' button
Step 1	System displays confirmation dialog: 'Are you sure you want to delete this alumni record? This action cannot be undone.'
Decision	Confirm Delete? If NO (Cancel): • Close dialog • Return to alumni list • No action taken If YES (Confirm): • Proceed to deletion process
Step 2	Check for related data: SELECT * FROM users WHERE alumni_id = ?
Check	Has User Account? If YES: • Delete user account first (cascade delete) • DELETE FROM users WHERE alumni_id = ? If NO: • Proceed directly to alumni deletion
Step 3	Delete photo file from storage: Get photo path from alumni record Delete file from filesystem or S3 bucket
Step 4	Delete alumni record from database: DELETE FROM alumni WHERE id = ?
Success	Display success message: 'Alumni record deleted successfully' Update dashboard statistics (decrease count) Refresh alumni list (remove deleted record) Redirect to alumni list

06. Conclusion

These flowcharts provide a comprehensive visual representation of the Alumni Web Application's functionality and data flow. They serve as essential documentation for development, testing, training, and maintenance purposes.

6.1 Key Takeaways

- **Three-Tier Architecture:**

Clear separation between client, API, and database layers ensures maintainability and scalability

- **Robust Security:**

Multiple security layers including bcrypt password hashing, JWT authentication, rate limiting, and HTTPS encryption

- **Role-Based Access:**

Clear distinction between admin full-access and user view-only permissions

- **Complete CRUD Operations:**

Administrators have full control over alumni data with proper validation and error handling

- **Data Integrity:**

Comprehensive validation at multiple levels ensures clean and accurate data

6.2 Usage Guidelines

For Developers:

- Use these flowcharts as reference during development
- Ensure all decision points and validations are properly implemented
- Follow the data flow patterns for consistent implementation

For Testers:

- Create test cases based on each flow path
- Verify all decision points handle both positive and negative scenarios
- Test error handling and validation messages

For Administrators:

- Understand the complete workflow of the system
- Reference these flowcharts during training sessions
- Use as troubleshooting guide when issues arise

These flowcharts will be updated as the system evolves to reflect new features and modifications.

Contact Information

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